

Study: Copper Price Forecasting - Model Evaluations

In the attached code, I ran a study to compare the accuracy of three different predictive models on a time series dataset of copper (physical) prices. The models I chose to explore were the ARIMA (AutoRegressive Integrated Moving Average) model, the SARIMA (Seasonal AutoRegressive Integrated Moving Average) model, and Meta's "Prophet" forecasting model.

The former two models are widely used in various fields such as economics, finance, and meteorology, and were the fundamental time series prediction methods that informed systematic trading strategies at the conception of the sub-industry before machine learning and neural networks grew in use.

"Prophet" is a relatively new model compared to the prior mentioned models having been developed by Meta Data Scientists in 2017. "Prophet" is known for its ease of use, automatic handling of seasonality, and its ability to effectively handle outliers ensuring more reliable forecasts.

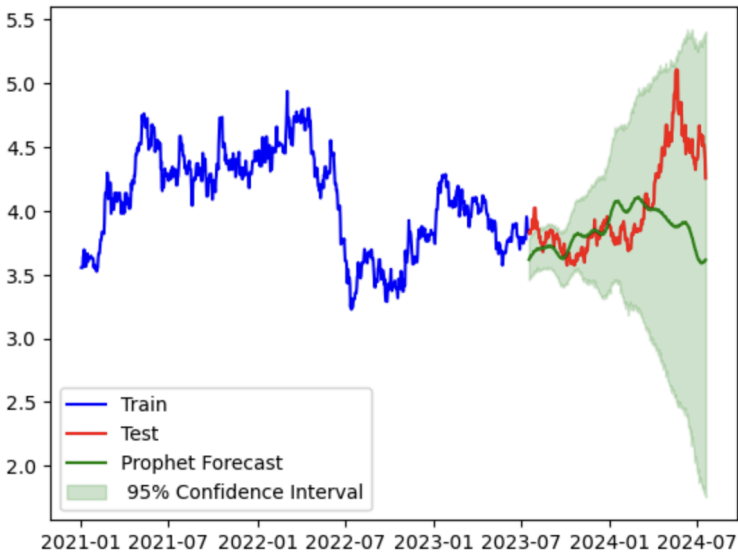
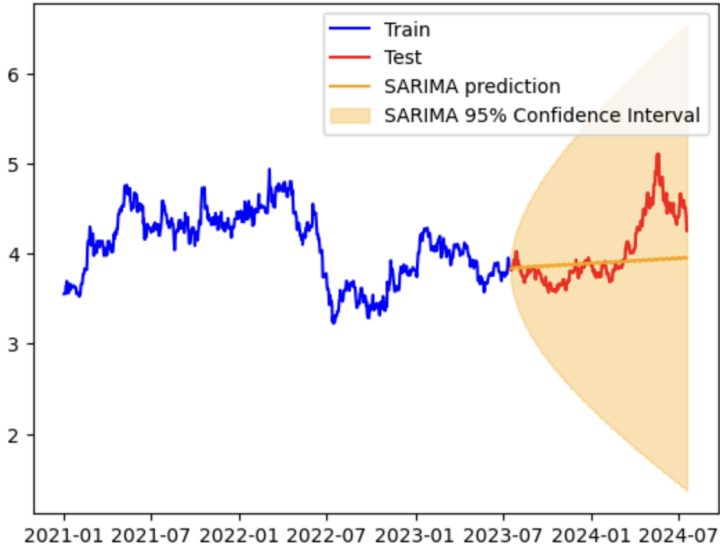
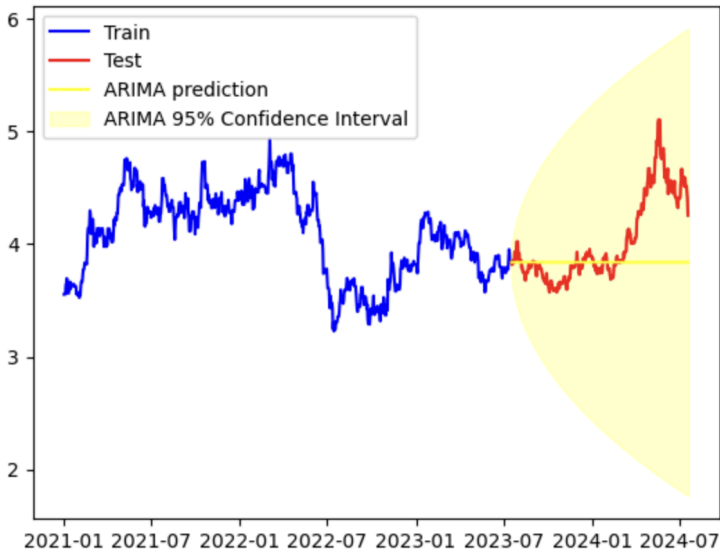
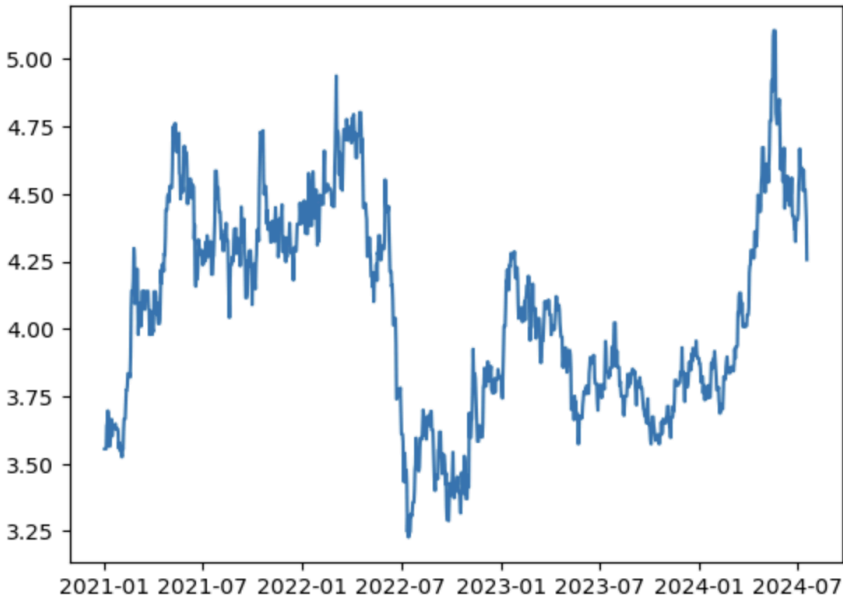
Of the three models used, the SARIMA model proved to have the highest predictive accuracy as determined by comparing the root mean squared error of each model's predictive output versus the actual observed data points in the time series' "test" set.

If I were to use this model to inform my trading strategy, I would fit the model to predict values for future dates and deploy a dynamic trailing limit to add or trim from my existing position when the price of copper deviates approximately one standard deviation away from its predicted value, and a dynamic protective stop-loss set to trigger when the price of copper moves beyond the model's 95% confidence interval band.

In the future, I would be interested in comparing the output of the autoregressive models to the output of a LSTM (Long-Short-Term-Memory) model, a type of neural network well-suited for time-series data that is especially helpful in capturing complex patterns and long-term dependencies. LSTM models are more complex than traditional time series models, and require careful tuning of hyperparameters and more computational resources. As a result of their complexity, LSTM models can be less interpretable than traditional autoregressive models.

Appendix

	value
count	1295.000000
mean	4.065911
std	0.381225
min	3.227200
25%	3.770550
50%	4.052800
75%	4.369700
max	5.105600



Model	RMSE
ARIMA	0.415121
SARIMA	0.368169
Prophet	0.426226